

TABLE 5-6 Control Functions and Microoperations for the Basic Computer

Fetch	$R'T_0$:	$AR \leftarrow PC$	
	$R'T_1$:	$IR \leftarrow M[AR], PC \leftarrow PC + 1$	
Decode	$R'T_2$:	$D_0, \dots, D_7 \leftarrow \text{Decode } IR(12-14),$ $AR \leftarrow IR(0-11), I \leftarrow IR(15)$	load AR=1
Indirect	D_7IT_3 :	$AR \leftarrow M[AR]$	دور 5، AR = قيمة
Interrupt:	$T_0T_1T_2(IEN)(FGI + FGO)$:	$R \leftarrow 1$	clear, increment
	RT_0 :	$AR \leftarrow 0, TR \leftarrow PC$	
	RT_1 :	$M[AR] \leftarrow TR, PC \leftarrow 0$	
	RT_2 :	$PC \leftarrow PC + 1, IEN \leftarrow 0, R \leftarrow 0, SC \leftarrow 0$	
Memory-reference:			
AND	D_0T_4 :	$DR \leftarrow M[AR]$	
	D_0T_5 :	$AC \leftarrow AC \wedge DR, SC \leftarrow 0$	
ADD	D_1T_4 :	$DR \leftarrow M[AR]$	
	D_1T_5 :	$AC \leftarrow AC + DR, E \leftarrow C_{out}, SC \leftarrow 0$	$LD(AR) =$ $R'T_0 + RT_2 + D_7IT_3$
LDA	D_2T_4 :	$DR \leftarrow M[AR]$	
	D_2T_5 :	$AC \leftarrow DR, SC \leftarrow 0$	بالقيمة التي في الذاكرة
STA	D_3T_4 :	$M[AR] \leftarrow AC, SC \leftarrow 0$	
BUN	D_4T_4 :	$PC \leftarrow AR, SC \leftarrow 0$	
BSA	D_5T_4 :	$M[AR] \leftarrow PC, AR \leftarrow AR + 1$	
	D_5T_5 :	$PC \leftarrow AR, SC \leftarrow 0$	inc PC
ISZ	D_6T_4 :	$DR \leftarrow M[AR]$	
	D_6T_5 :	$DR \leftarrow DR + 1$	
	D_6T_6 :	$M[AR] \leftarrow DR, \text{ if } (DR = 0) \text{ then } (PC \leftarrow PC + 1), SC \leftarrow 0$	
Register-reference:			
	$D_7I'T_3 = r$	(common to all register-reference instructions)	
	$IR(i) = B_i$	(i = 0, 1, 2, ..., 11)	
	r:	$SC \leftarrow 0$	تتم القراءة
CLA	rB_{11} :	$AC \leftarrow 0$	القيمة ← M[AR]
CLE	rB_{10} :	$E \leftarrow 0$	
CMA	rB_9 :	$AC \leftarrow \overline{AC}$	
CME	rB_8 :	$E \leftarrow \overline{E}$	clear SC
CIR	rB_7 :	$AC \leftarrow \text{shr } AC, AC(15) \leftarrow E, E \leftarrow AC(0)$	OK → IEN
CIL	rB_6 :	$AC \leftarrow \text{shl } AC, AC(0) \leftarrow E, E \leftarrow AC(15)$	
INC	rB_5 :	$AC \leftarrow AC + 1$	
SPA	rB_4 :	If $(AC(15) = 0)$ then $(PC \leftarrow PC + 1)$	
SNA	rB_3 :	If $(AC(15) = 1)$ then $(PC \leftarrow PC + 1)$	
SZA	rB_2 :	If $(AC = 0)$ then $PC \leftarrow PC + 1$	
SZE	rB_1 :	If $(E = 0)$ then $(PC \leftarrow PC + 1)$	
HLT	rB_0 :	$S \leftarrow 0$	
Input-output:			
	$D_7IT_3 = p$	(common to all input-output instructions)	
	$IR(i) = B_i$	(i = 6, 7, 8, 9, 10, 11)	
	p:	$SC \leftarrow 0$	
INP	pB_{11} :	$AC(0-7) \leftarrow INPR, FGI \leftarrow 0$	
OUT	pB_{10} :	$OUTR \leftarrow AC(0-7), FGO \leftarrow 0$	
SKI	pB_9 :	If $(FGI = 1)$ then $(PC \leftarrow PC + 1)$	
SKO	pB_8 :	If $(FGO = 1)$ then $(PC \leftarrow PC + 1)$	
ION	pB_7 :	$IEN \leftarrow 1$	
IOF	pB_6 :	$IEN \leftarrow 0$	

بایستی bus

$$x_0: \emptyset \quad \overline{AR} \quad x_1: \emptyset \quad \overline{PC} \quad x_2: T_0 \quad \overline{DR} \quad x_3: D_5 T_6 + D_2 T_5 \quad \overline{AC} \quad x_4: \emptyset$$

$$\overline{IR} \quad x_5: T_2 \quad \overline{TR} \quad x_6: \emptyset \quad \overline{RAM} \quad x_7: T_1 + D_7' I T_3 + D_1 T_4 + D_2 T_4 + D_6 T_4$$

x_i ها درونی encoder اند و بعد از مدتی S_0, S_1, S_2 از بیرون می آیند.

بایستی (load, inc, clr) register

$$\overline{AR} \quad \text{load: } T_0 + T_2 + D_7' I T_3 \quad \text{inc: } \emptyset \quad \text{clr: } \emptyset$$

$$\overline{PC} \quad \text{load: } \emptyset \quad \text{inc: } T_1 + D_6 T_6 \quad \text{clr: } \emptyset$$

$(\overline{DR} = \emptyset)$
 $(DR(0) + \dots + DR(8))'$

$$\overline{DR} \quad \text{load: } D_1 T_4 + D_2 T_4 + D_6 T_4 \quad \text{inc: } D_6 T_5 \quad \text{clr: } \emptyset$$

$$\overline{AC} \quad \text{load: } D_1 T_5 + D_2 T_5 \quad \text{inc: } \emptyset \quad \text{clr: } \emptyset$$

$$\overline{IR} \quad \text{load: } T_1 \quad \text{inc: } \emptyset \quad \text{clr: } \emptyset$$

$$\overline{TR} \quad \text{load: } \emptyset \quad \text{inc: } \emptyset \quad \text{clr: } \emptyset$$

بایستی RAM (read, write)

$$\overline{\text{read}} \quad T_1 + D_7' I T_3 + D_1 T_4 + D_2 T_4 + D_6 T_4$$

$$\overline{\text{write}} \quad D_6 T_6$$

نسل flip flop

[E] $D_1 T_5 : (\text{load count})$

[S] $\underbrace{(D_7' I T_3)}_F B_0 : \text{clr } S$

نسل sequence counter

$D_1 T_5 + D_2 T_5 + D_6 T_6 + D_7' I T_3 B_0 : \boxed{\text{clr } SC}$

در بقیه موارد فقط inc

①

0	0010	0011	LDA
1	0001	0100	ADD
2	0111	0001	HLT

3	1000	0000	X_1
4	1000	0001	X_2

②

0	I	1110	0011	ISZ
2		0111	0001	HLT

3		0000	0111	X_1
7		1111	1111	X_2